

# Assignment 1: Network Latencies, Ping, and Traceroute

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## Repository

The complete source code, input data, and instructions needed to reproduce the experiments and plots are available at:

<https://github.com/RohMalpani/CS422Assignments>

Reproduction command:

```
./run_all.sh
```

## 1 Ping Test and Round-Trip Time

### 1.1 Method

We tested every destination listed in the provided IP-address file, along with our own public IP address. The script sent ten ICMP echo requests to each destination and recorded the minimum, average, and maximum RTT, using NA for non-responsive destinations. It obtained latitude and longitude from [ip-api.com](https://ip-api.com) and calculated great-circle distance with the Haversine formula using an Earth radius of 6,371 km.

The code used to collect and process the ping measurements is available here: [ping collection](#) and [RTT calculation](#).

### 1.2 Results

The first five rows are shown below (showing 5/37 results only). The complete data are available in `part1/results.csv`.

Table 1: First five ping results.

| Destination IP  | Distance (km) | Min RTT (ms) | Avg RTT (ms) | Max RTT (ms) |
|-----------------|---------------|--------------|--------------|--------------|
| 52.119.103.17   | 0             | 4.209        | 12.189       | 56.938       |
| 160.242.19.254  | 11,498        | 298.388      | 319.136      | 367.114      |
| 41.110.39.130   | 7,463         | 164.185      | 182.277      | 246.594      |
| 213.158.175.240 | 9,918         | 168.356      | 222.082      | 356.200      |
| 102.214.66.39   | 9,327         | 211.874      | 248.932      | 322.786      |

### 1.3 Distance versus RTT

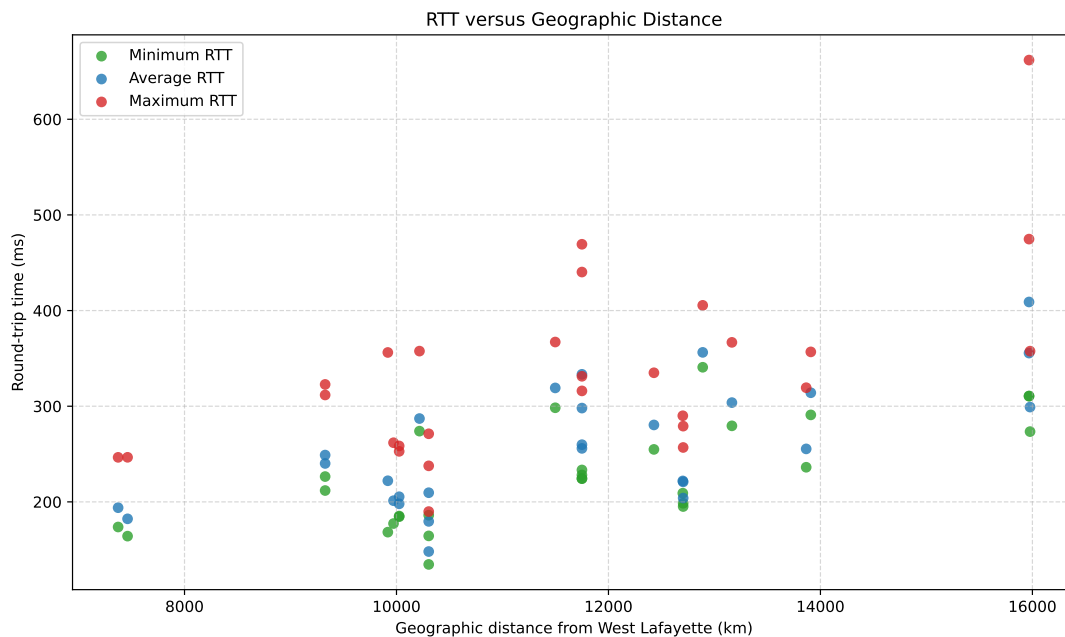


Figure 1: Geographic distance from West Lafayette, Indiana, versus round-trip time. Each point represents one destination IP address.

The plotting code is available here: [distance-versus-RTT plot](#).

### 1.4 Discussion

RTT generally increases with geographic distance because signals need more time to travel over longer physical paths. The relationship is not exact, however, because packets follow routes selected by network providers rather than a direct geographic path. Peering locations, indirect routing, and router processing can all add latency.

The minimum RTT is the best estimate of the path's baseline latency. It mostly reflects propagation, transmission, and processing delay with little temporary queueing. The maximum RTT is the slowest observed reply and shows how network conditions varied during the measurement. Congestion, queueing, route changes, and ICMP processing priority can increase maximum RTT above the baseline. Therefore, distance establishes an important lower bound on RTT, while routing and short-term network conditions explain much of the variation.

## 2 Latency Breakdown

### 2.1 Method

Five destinations were selected uniformly at random from the iperf3 server list using seed 42.

Table 2: Five destinations selected for traceroute analysis.

| Destination IP    | Site/Country     | Outcome          |
|-------------------|------------------|------------------|
| 185.152.67.2      | Los Angeles, US  | Reached, 12 hops |
| 89.187.160.1      | Tokyo, JP        | Reached, 12 hops |
| 105.235.237.2     | Bata, GQ         | Reached, 21 hops |
| 103.146.200.98    | Port Moresby, PG | Reached, 21 hops |
| speed2.fiberby.dk | Copenhagen, DK   | Reached, 14 hops |

Each destination was traced in ICMP-echo mode with three probes per hop, a two-second timeout, and a 30-hop limit. The per-hop RTT is the minimum of the responsive probes. Non-responsive hops are recorded as \*, and a destination that does not respond after a retry is skipped.

The traceroute collection and parsing code is available here: [traceroute invocation, parsing, and reachability check](#), with the retry/skip logic here: [batch runner with retry-once-before-skip](#). Target selection is here: [seeded random target selection](#).

## 2.2 Per-Hop Latency Breakdown

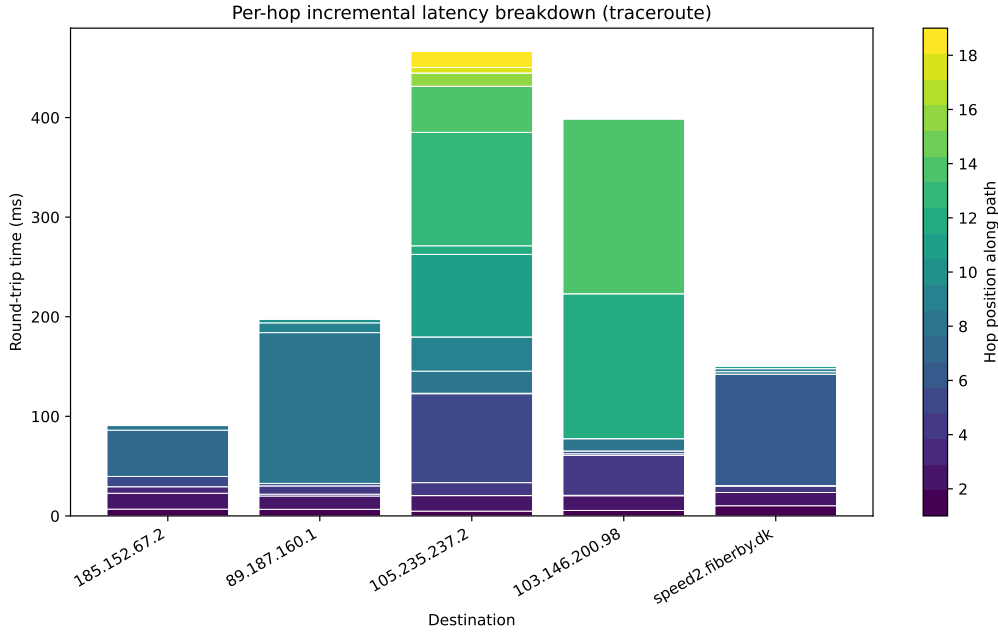


Figure 2: Stacked latency breakdown for the responsive hops along the paths to the five selected destinations.

The plotting code is available here: [stacked per-hop latency chart](#), including how incremental (not cumulative) latency is computed: [incremental\\_latencies\(\)](#).

## 2.3 Hop Count versus RTT

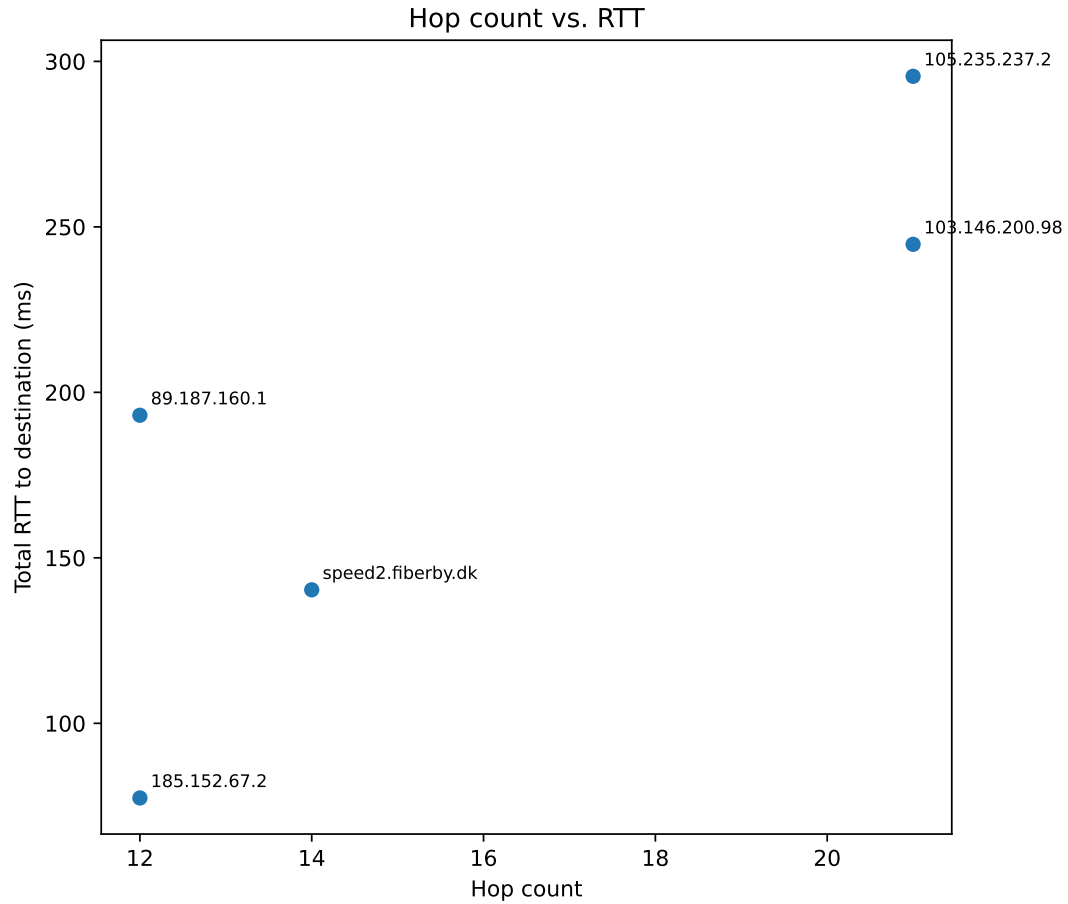


Figure 3: Hop count versus destination RTT. Each point represents one of the five selected destination IP addresses.

The plotting code is available here: [hop-count-versus-RTT plot](#).

## 2.4 Discussion

The stacked bar chart shows that latency is concentrated in a few larger increments, usually in the long-haul part of the route, rather than being distributed evenly across hops. Non-responsive hops have no measured RTT and are omitted from the breakdown.

Hop count has only a weak positive relationship with RTT. Routes with the same number of hops can have different RTTs because the distance and quality of their links differ.